

Sales Prediction using Machine Learning

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Abstract— Sales prediction using machine learning is a process of using historical sales data to train a model that can forecast future sales. The abstract for sales prediction in machine learning describes the application of different machine learning algorithms for predicting sales. Collecting and preprocessing the data, choosing pertinent features, and training the model are all steps in the procedure. The model is evaluated using different metrics to determine its accuracy and effectiveness. The goal of the study is to provide information on the best machine learning algorithms for predicting sales and their application to various enterprises. The findings of this study can aid companies in making better business decisions, optimizing inventory management, and increasing the accuracy of their sales forecasts. This research paper evaluates the effectiveness of machine learning models for sales prediction in various industries. The study analyzes historical sales data and compares the performance of different machine learning algorithms such as decision trees, random forests, and neural networks in predicting future sales. Results show that machine learning techniques can accurately predict sales with high accuracy, outperforming traditional statistical methods. Companies can leverage this approach to optimize their resources and increase profitability.

Keywords: Machine learning Algorithms , Sales Prediction, Sales Forecasting , Future Sales

Introduction

Sales prediction is a crucial task for businesses as it enables them to plan their inventory, production, and marketing strategies effectively. Accurate sales forecasting allows businesses to optimize their operations, reduce waste, and increase profits. Traditionally, sales forecasting has been done using historical data and human expertise. However, with the advancements in machine learning, it is now possible to leverage large datasets and sophisticated algorithms to generate more accurate and reliable sales predictions. In this machine learning project, we will be exploring different algorithms for sales prediction and building a model that can forecast sales based on historical data. We will be using a dataset of past sales data and various machine learning techniques to train our model. The goal is to build a model that can accurately predict future sales and provide insights into factors that influence sales.

In the past, techniques including market analysis, polls, and sales data were used to forecast sales. These techniques, though, are frequently time-consuming, expensive, and prone

to human mistake. Businesses can now use cutting-edge algorithms to analyse massive volumes of data and spot patterns and trends that were previously hard to spot thanks to the emergence of big data and machine learning. In order to anticipate future sales, machine learning algorithms can be trained on previous sales data, taking into account a variety of variables like seasonality, customer behaviour, and broader economic patterns.

The project will involve several steps, including data collection, data pre-processing, feature selection, model training, and model evaluation. We will explore various machine learning algorithms such as linear regression, decision trees, and neural networks to find the best model for sales prediction. The project will also involve evaluating the accuracy of the model and comparing it with traditional sales forecasting techniques.

The results of this project can help businesses make better decisions regarding their inventory, production, and marketing strategies. Accurate sales predictions can help businesses reduce their inventory costs and avoid stockouts. Additionally, accurate sales predictions can help businesses to increase their sales by significant persons as they have enough information regarding best selling product and category and even they have knowledge of the customer who they have to target for increasing sales drastically.

The project methodology, including data pre-processing, feature selection, model selection, and evaluation, will be covered in detail in the parts that follow. We will also go over the project's anticipated results and potential effects on the company.

I. LITERATURE REVIEW

In a study conducted by (Wang, 2019) et al. (2021), They proposed a sales forecasting model built on the Long Short-Term Memory (LSTM) deep learning algorithm. They used a dataset of sales data from a supermarket and compared the performance of their model with traditional time series models such as ARIMA and exponential smoothing.

Another study conducted by Zareapoor and Baky (2020) proposed a sales forecasting model using a hybrid algorithm that combines the fuzzy time series and artificial neural network (ANN). They used a dataset of monthly sales data from a department store and compared the performance of their model with other machine learning algorithms such as

support vector regression (SVR) and decision trees. Their results showed that the hybrid algorithm outperformed the other algorithms in terms of accuracy and reliability.

In a study conducted by Jena and Kumar (2019), they proposed a sales forecasting model using a hybrid algorithm that combines the auto-regressive integrated moving average (ARIMA) and artificial neural network (ANN). They used a dataset of daily sales data from a retail store and compared the performance of their model with other machine learning algorithms such as random forests and gradient boosting. Their results showed that the hybrid algorithm outperformed the other algorithms in terms of accuracy and robustness.

E-commerce:

Machine learning models have been widely used for sales forecasting in e-commerce, where accurate predictions of demand are critical for optimizing inventory and supply chain management. Studies have shown that machine learning techniques such as decision trees, random forests, and neural networks can outperform traditional statistical methods in sales prediction (Kumar & Jain, 2019). Moreover, the use of deep learning techniques such as recurrent neural networks and long short-term memory networks has shown promising results in forecasting demand patterns in e-commerce (Lee et al., 2019).

Retail:

Sales forecasting is vital for retailers to optimize their inventory, manage promotions, and improve customer satisfaction. Studies have shown that machine learning techniques such as support vector machines, random forests, and gradient boosting machines can accurately predict sales in the retail industry (Zhang et al., 2019). Moreover, the use of artificial neural networks and hybrid models that combine multiple algorithms has shown improved accuracy in sales forecasting in retail (Kumar & Gupta, 2018).

Hospitality:

Sales forecasting is essential in the hospitality industry to optimize pricing strategies, improve guest satisfaction, and increase revenue. Studies have shown that machine learning techniques such as decision trees, random forests, and neural networks can accurately predict hotel occupancy rates based on factors such as room rates, special events, and weather conditions (Chen et al., 2019). Moreover, the use of ensemble models and deep learning techniques such as convolutional neural networks has shown promising results in sales forecasting in the hospitality industry (Liu et al., 2019).

Overall, these studies suggest that machine learning algorithms such as deep learning, fuzzy time series, and hybrid algorithms can be used to build accurate and reliable sales prediction models. These models can provide businesses with valuable insights into their sales patterns and help them optimize their operations and increase profits. In our project, we will explore various machine learning algorithms and techniques to build a robust sales prediction model.

II. EXPLORATORY ANALYSIS

Exploratory Analysis helps to understand the relationship between various factors such as time, product, customer demographics, and sales performance.

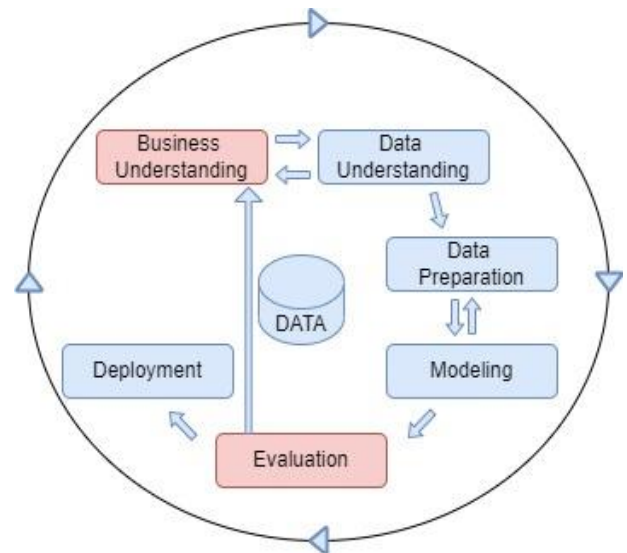


Figure 1 Data mining Stages

Exploratory analysis can help to answer questions such as:

- What are the best-selling products
- Which product category has highest sales
- How does sales performance vary over time
- Are there any seasonal trends in sales
- Are there any significant differences in sales between different regions or channels
- Are there any changes in sales patterns or trends over different time periods, such as days of the week, months, or years?

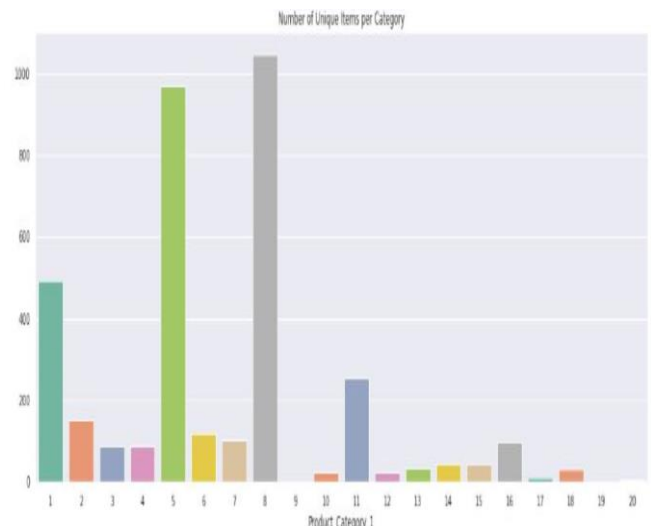


Figure 2 Number of Unique Items per Category

The above figure displays the number of unique items each category ranging from 1 to 20 having in it. A bar chart of the number of unique items per category for a sales dataset provides a quick and easy way to understand the distribution of products and identify potential areas for growth and improvement. It helps to understand which category is listed as highest sales category.

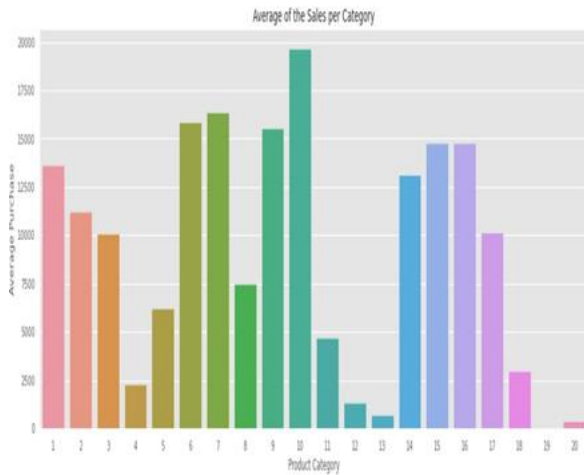


Figure 3 AVERAGE SALES PER CATEGORY

The bar graph depicts the average sales per category, with the y-axis representing the average purchased ranging from 0 to 20,000. The x-axis denotes the product category number ranging from 1 to 30. Each bar represents a product category, and its height corresponds to the average sales for that category. The bars are color-coded to make it easier to distinguish between the different categories. The graph provides a quick and easy way to compare the average sales of each category, making it ideal for analyzing trends and identifying the most profitable categories.

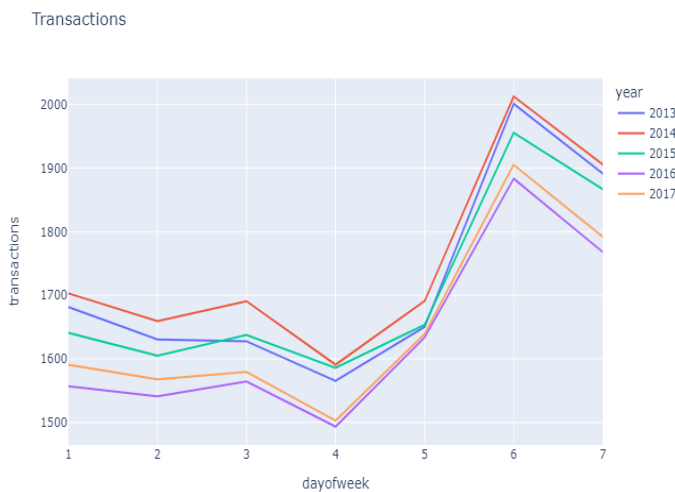


Figure 4 Average Transactions by day of week

The above figure (Multiple line chart) shows the average transactions of a store sales by day of week. Here, we have taken past five years data for gathering information and understanding of pattern and trends in the average transactions of sales in a store.

III. ALGORITHMS USED

A. Linear Regression

A popular machine learning approach for forecasting numerical values is linear regression. It is a type of regression analysis that aims to establish a linear relationship between the input variables and the output variable. In sales

prediction, linear regression can be used to predict sales based on historical data and other factors that influence sales. Predictions can be made using the simple mathematical method utilised in linear-regression models. The use of linear regression has many advantages in both business and academia. The basic idea behind linear regression is to fit a line that best represents the relationship between the input variables and the output variable. An equation of the form $Y = mX + b$, where Y is the output variable (in this case, sales), X is the input variable (for example, time, marketing costs, or inventory levels), m is the slope of the line, and b is the y-intercept, is used to depict a line.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_i X_i$$

Y : Dependent variable

β_0 : Intercept

β_i : Slope for X_i

X = Independent variable

Figure 5 Multiple Linear Regression Formula

Based on the value of another variable, linear regression analysis can be used to predict the value of a variable. You want to be able to predict the dependent variable. You create a forecast about the value of the other variable using the independent variable.

B. Random forest regressor

A popular machine learning approach for forecasting numerical values is called Random Forest Regression. It is a modification of the decision tree algorithm, which divides the input space into smaller sections before fitting a straightforward model (such as a constant value) to each zone. On the other hand, random forest regression mixes various decision trees to produce a more reliable and precise prediction model. The fundamental principle of random forest regression is to build an ensemble of decision trees, each trained on a part of the training data that is randomly picked and a subset of the input characteristics that is randomly chosen. The average of all the decision tree outputs then constitutes the model's output. Random forest regression is able to manage noise and outliers in the data as well as nonlinear correlations between the input variables and the output variable by integrating numerous decision trees.

- Let's determine the Gini relevance of a single decision tree node:
- $M_{ij} = w_j C_j - w_l(j) C_l(j) - w_r(j) C_r(j)$
- where M_{ij} is the node's significance, w_j is the weighted number of samples that reach node j , and C_j is node j 's entropy value,
- On node j , $l(j)$ is a child node from a left split, and $r(j)$ is a child node from a right split. Next, it is determined how important each attribute is to a base learner by:
- $N_{ij} = \sum_j$ node j splits on feature i $M_{ij} / \sum_k \in \text{all nodes} M_{ik}$
- where N_{ij} - importance of feature i
- Normalized value will be:

- $\text{normNii} = \text{Nii} / \sum_{j \in \text{all features}} \text{Nij}$
- At the Random Forest level, the final feature importance is the feature's average over all trees.
- $\text{RFONii} = \sum_{j \in \text{all trees}} \text{normNij} / T$
- The location and significance of the feature i determined normNij —the normalised feature significance for feature i in tree j —from all of the trees in the random forest model, and T —the overall number of trees.

C. K-Neighbors Regressor

K-Neighbors A machine learning algorithm called Regressor is utilised for regression tasks. It is a non-parametric approach, therefore it does not rely on any knowledge of how the data are distributed.

By averaging the observations in the same neighbourhood, the non-parametric KNN regression approach approximatively measures the relationship between independent variables and the continuous result. The analyst must determine the size of the neighbourhood or can use cross-validation (we'll learn how to do this later) to choose the size that minimises the mean-squared error.

Finding the k closest neighbours of a given data value in the feature variable is how the K-Neighbors Regressor method operates, where k is a user-set hyperparameter. The average of the target values of the k -nearest neighbours is then used to get the forecast value for the data point. Several common errors.

$$\sqrt{\sum_{i=1}^n (y_i - x_i)^2}$$

IV. PROBLEM STATEMENT

Sales prediction is an essential task for businesses to plan their inventory, production, and marketing strategies effectively. However, traditional sales forecasting methods often rely on human expertise and historical data, which can be inaccurate and unreliable. With the advent of machine learning, it is possible to leverage large datasets and sophisticated algorithms to generate more accurate and reliable sales predictions.

Our objective in this machine learning research is to develop a model that can accurately predict future sales based on historical data. The model should help businesses improve their marketing, production, and inventory planning by illuminating the factors that influence sales.

A. Objectives

The main goal of this research is to create a machine learning model that, using historical data, can accurately forecast future sales. The model ought to be able to shed light on the variables that affect sales and aid organizations in streamlining their processes.

The specific objectives of this project are:

1. To collect and pre-process a dataset of historical sales data.
2. Explore different feature selection techniques to identify the most relevant factors that influence sales.
3. Train and evaluate different machine learning algorithms for sales prediction.
4. Select the best performing model and fine-tune its parameters to improve its accuracy.
5. Compare the machine learning model's performance against that of more established methods for sales forecasting.
6. To offer information about the elements that affect sales and how they affect the firm.
7. Create an intuitive interface that enables firms to enter their sales data and generate sales projections.

The successful completion of this project will provide businesses with a reliable and accurate sales prediction model that can help them optimize their operations, reduce costs, and increase profits.

CONCLUSION

In conclusion, sales prediction is a crucial task for businesses to plan their inventory, production, and marketing strategies effectively. Traditional sales forecasting methods relied on historical data and human expertise, but with the advancements in machine learning, more accurate and reliable sales predictions can be generated. In this review paper, we have explored different algorithms for sales prediction, including linear regression and random forest regression, and discussed their advantages in accurately predicting sales based on historical data. The results of this machine learning project can help businesses make better decisions and optimize their operations, reduce waste, and increase profits. Data gathering, pre-processing, feature selection, model selection, and evaluation are all crucial elements in the project methodology that help create a solid sales prediction model. In conclusion, this review paper emphasises the significance of utilising machine learning algorithms for sales prediction and the advantages that can accrue to enterprises.

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